1. Algebra and functions		
1.1	Laws of indices for all rational exponents.	$a^m \times a^n = a^{m+n}$, $a^m \div a^n = a^{m-n}$, $(a^m)^n = a^{mn}$ The equivalence of $a^{\frac{m}{n}}$ and $\sqrt[n]{a^m}$ should be known.
1.2	Use and manipulation of surds.	Students should be able to rationalise denominators.
1.3	Quadratic functions and their graphs.	
1.4	The discriminant of a quadratic function.	Need to know and to use $b^2 - 4ac > 0$, $b^2 - 4ac = 0$ and $b^2 - 4ac < 0$
1.5	Completing the square. Solution of quadratic equations.	Solution of quadratic equations by factorisation, use of the formula, use of a calculator and completing the square. $ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2 + \left(c - \frac{b^2}{4a}\right)$
1.6	Solve simultaneous equations; analytical solution by substitution.	
1.7	Interpret linear and quadratic inequalities graphically.	For example, $ax + b > cx + d$, $px^2 + qx + r \geqslant 0$, $px^2 + qx + r < ax + b$. Interpreting the third inequality as the range of x for which the curve $y = px^2 + qx + r$ is below the line with equation $y = ax + b$. Including inequalities with brackets and fractions. These would be reducible to linear or quadratic inequalities, e.g. $\frac{a}{x} < b$ becomes $ax < bx^2$, $x \neq 0$.
1.8	Represent linear and quadratic inequalities graphically.	Represent linear and quadratic inequalities such as $y > x + r$ and $y > ax^2 + bx + c$ graphically. Shading and use of dotted and solid line convention is required.
1.9	Solutions of linear and quadratic inequalities.	For example, solving $ax + b > cx + d$, $px^2 + qx + r \geqslant 0$, $px^2 + qx + r < ax + b$.
1.10	Algebraic manipulation of polynomials, including expanding brackets and collecting like terms, factorisation.	Students should be able to use brackets. Factorisation of polynomials of degree n, $n \leqslant 3$, e.g. $x^3 + 4x^2 + 3x$. The notation $f(x)$ may be used.

What students need to learn:		Guidance
1. Algebra and functions *continued*		
1.11	Graphs of functions; sketching curves defined by simple equations. Geometrical interpretation of algebraic solution of equations. Use of intersection points of graphs of functions to solve equations.	Functions to include simple cubic functions and the reciprocal functions $y = \frac{k}{x}$ and $y = \frac{k}{x^2}$ with $x \neq 0$. Knowledge of the term asymptote is expected. Also, trigonometric graphs.
1.12	Knowledge of the effect of simple transformations on the graph of $y = \mathrm{f}(x)$ as represented by $y = a\mathrm{f}(x)$, $y = \mathrm{f}(x) + a$, $y = \mathrm{f}(x + a)$, $y = \mathrm{f}(ax)$.	Students should be able to apply one of these transformations to any of the above functions (quadratics, cubics, reciprocals, sine, cosine, and tangent) and sketch the resulting graphs. Given the graph of any function $y = \mathrm{f}(x)$, students should be able to sketch the graph resulting from one of these transformations.